

L2 DATA PLATFORM SUPPORT

Interview Preparation Guide

Snowflake · dbt · Apache Airflow

SLA Management · Incident Response · Behavioral · Communication Templates

What Makes L2 Support Different

L2 Support engineers are tested on three dimensions simultaneously — technical depth, SLA awareness, and communication under pressure. Most candidates prepare for each separately and fail because they cannot combine them.

What Is Tested	What It Means	How to Demonstrate It
Technical depth	Diagnose a broken production system from first principles	Name specific tools. Query Profile, ACCOUNT_USAGE, SYSTEM dollar PIPE_STATUS. Show you know what to check first.
SLA awareness	Understand urgency. Know when to escalate versus solve.	Always mention: severity level, business impact, SLA timer elapsed, escalation threshold. Never say 'I would just fix it'.
Communication	Explain a Snowflake issue to a VP who does not know SQL	Translate every technical answer into a plain-English stakeholder update. Practice the two-sentence version of every root cause.

The L2 Mindset — Four Phases of Every Incident

Phase	Time Budget	What You Do	What You Communicate
Detect and Triage	0–5 min	Assess severity. Identify business impact. Check SLA timer.	Acknowledge: 'Received, investigating. Update in 15 min.'
Diagnose	5–20 min	Check logs, Query Profile, Airflow UI, dbt run output. Narrow root cause.	Status: 'Root cause identified as X. Working on fix.'
Contain	20–45 min	Stop the bleeding — pause pipeline, retry task, provide temporary workaround.	Update: 'Workaround in place. Full resolution ETA: X time.'
Resolve	45–90 min	Fix root cause, validate data integrity, confirm recovery.	Resolution: 'Issue resolved. Root cause was X. Preventive action: Y.'

IMPORTANT

SLA timers — know these cold:

P1 Critical: 15-min response, 2-hr resolution

P2 High: 30-min response, 4-hr resolution

P3 Medium: 4-hr response, 24-hr resolution

CHAPTER 1

Snowflake — L2 Troubleshooting

Architecture · Performance · Access · Pipelines · Cost

Architecture Symptoms and First Checks

Symptom	Affected Layer	First Check	L2 Action
Query 10x slower than baseline	Query Processing (warehouse)	Query Profile — partition scan ratio and spill columns	Size up warehouse or add clustering key
Insufficient privileges error	Cloud Services (access control)	SHOW GRANTS TO ROLE — check all 4 required grants	Grant missing privilege, add FUTURE TABLES grant
Snowpipe not loading files	Cloud Services plus Storage	SYSTEM dollar PIPE_STATUS — executionState and notificationChannelStatus	Resume pipe, fix SQS or IAM, then REFRESH
Result cache not working	Cloud Services	Check query for CURRENT_TIMESTAMP, RANDOM, or UUID_STRING	Remove non-deterministic functions from query
Warehouse credit spike	Query Processing	WAREHOUSE_METERING_HISTORY — find idle warehouses	Set AUTO_SUSPEND = 60, add resource monitor

The Four Required Grants — Most Common L2 Ticket

Over 60% of access-denied tickets come from one missing grant. Every new user needs all four to run a SELECT:

- GRANT USAGE ON DATABASE — allows seeing the database
- GRANT USAGE ON SCHEMA — allows seeing the schema and its objects
- GRANT SELECT ON TABLE — allows reading the table data
- GRANT USAGE ON WAREHOUSE — allows using compute to run the query

TIP

Add GRANT SELECT ON FUTURE TABLES IN SCHEMA to prevent having to re-grant every time a new table is created.

Snowpipe Failure — Diagnosis Runbook

Step	Action	What to Look For
1	Run SYSTEM dollar PIPE_STATUS on the pipe name	executionState: RUNNING is healthy, PAUSED means manually paused
2	If paused: ALTER PIPE ... SET PIPE_EXECUTION_PAUSED = FALSE	Pipe should return to RUNNING state
3	Check INFORMATION_SCHEMA PIPE_USAGE_HISTORY	ERROR_COUNT and STATUS columns — look for LOAD_FAILED entries

4	Inspect SQS queue in AWS console	Dead-letter queue messages indicate S3 event notification issues
5	Run ALTER PIPE ... REFRESH MODIFIED_AFTER to catch up	Specify a timestamp 1–2 hours before the outage started
6	Verify with INFORMATION_SCHEMA COPY_HISTORY	FILE_NAME, ROW_COUNT, STATUS columns confirm files loaded

Root Cause	Signal	Fix
SQS disconnected	notificationChannelStatus = disconnected	Reconfigure S3 event notification with SQS ARN from SHOW PIPES
IAM role revoked	Auth error in pipe status; AccessDenied in CloudTrail	Fix IAM trust policy in AWS — add Snowflake IAM user ARN back
Pipe paused	executionState = PAUSED	ALTER PIPE SET PIPE_EXECUTION_PAUSED = FALSE then REFRESH
File format mismatch	ERROR_COUNT > 0 in load history	Fix file format or use ON_ERROR = CONTINUE; investigate bad files

Pipeline Task Failure Recovery

When an Airflow task fails mid-run, the Snowflake Stream offset does NOT advance because the consuming DML never committed. Stream data is preserved. You can safely retry without data loss.

Task State	What Happened	Recovery Action
FAILED	Task threw an exception	Check task log, fix root cause, then clear-and-rerun in Airflow UI
SKIPPED	Upstream task failed — this task was never started	Fix the failed upstream task, then clear from that task forward
ZOMBIE	Task appears running but worker process died	Kill zombie in Airflow UI, check worker health, requeue the task
QUEUED > 30 min	Pool is full or global concurrency limit hit	Check pool usage, increase pool slots or reduce concurrent tasks

Credit Spike Investigation — Sequence

- Step 1: Query `ACCOUNT_USAGE.METERING_DAILY_HISTORY` — find the exact day costs spiked
- Step 2: Query `WAREHOUSE_METERING_HISTORY` for that date — find which warehouse drove it
- Step 3: Check `WAREHOUSE_METERING_HISTORY LEFT JOIN QUERY_HISTORY` — find warehouses with credits but zero queries (idle billing)
- Step 4: Run `SHOW WAREHOUSES` — check `AUTO_SUSPEND` column; value 0 means never suspends
- Step 5: Immediate fix — `ALTER WAREHOUSE SET AUTO_SUSPEND = 60`
- Step 6: Add resource monitor with `NOTIFY` at 75%, `SUSPEND` at 100%

CHAPTER 2

dbt — L2 Troubleshooting

Model failures · Tests · Selectors · Snowflake integration

dbt Model Failure Diagnosis — Decision Tree

Error Type	How to Identify	First Action	Common Root Cause
Compilation error	Error before SQL runs — Jinja or ref() issue	Run dbt compile --select model — check target/compiled output	Typo in ref(), wrong variable name, circular dependency
Run-time error	SQL fails in Snowflake — model compiled but failed to execute	Copy compiled SQL from dbt log and run directly in Snowflake	Schema drift, type mismatch, NULL in non-nullable, permission issue
Test failure	Model ran but test returned rows	Run the test query manually — SELECT the failing rows	Source data quality issue, incremental filter bug, wrong accepted_values
Incremental needs full-refresh	column X of relation does not exist	Run dbt run --select model --full-refresh	Column added or removed from source since incremental was created

dbt Selector Quick Reference for L2

Selector	Runs	When to Use
--select model_name	Just that model	Test one specific broken model
--select model_name+	Model and all downstream	Check blast radius of a change
--select +model_name	Model and all upstream dependencies	Rebuild the full dependency chain
--select tag:daily	All models tagged daily	Run only the daily refresh set
--select path:models/staging	All models in staging folder	Run one entire layer
--select state:modified+	Changed models plus downstream	CI/CD — only rebuild what changed
--exclude tag:deprecated	Everything except deprecated models	Skip legacy models
--full-refresh with --select	Force drop-and-recreate the specified model	Fix broken incremental after schema change

profiles.yml — Snowflake Connection Errors

Error	Root Cause	Fix
Account not found	Wrong account identifier format (should be org-accountname)	Check Snowsight URL for correct format; org-account or legacy mycompany.us-east-1
JWT token is invalid	Private key file corrupt or wrong passphrase	Regenerate RSA key pair; update Snowflake user with new public key

Role does not exist	Role name case mismatch (Snowflake roles are case-sensitive)	Run SHOW ROLES in Snowflake; fix exact name in profiles.yml
Warehouse does not exist	Wrong warehouse name in profile	Run SHOW WAREHOUSES; copy exact name to profiles.yml

CHAPTER 3

Airflow — L2 Troubleshooting

Task states · SLA misses · Connections · Scaling

Task States — Every L2 Must Know

State	Meaning	Common Cause	L2 Action
running	Currently executing	Normal — monitor for SLA breach	Check duration vs SLA; check Query Profile if slow
success	Completed successfully	Normal	No action
failed	Exception thrown	Code error, connection issue, resource limit	Read task log; fix root cause; clear-and-rerun
skipped	Intentionally skipped	BranchOperator chose another path	Expected — verify branch logic is correct
upstream_failed	Upstream failed — task never started	Dependency failed before this task could run	Fix upstream failure; clear from that task forward
zombie	Appears running but worker died	Worker OOM, pod crash, network loss	Kill zombie in UI; check worker; requeue
queued	Waiting in scheduler queue	Workers busy, pool full, concurrency limit	Check pool slots; check global PARALLELISM setting

SLA Miss vs Task Failure — Critical Distinction

This is one of the most common L2 interview questions. SLA miss and task failure are entirely separate events in Airflow.

Concept	SLA Miss	Task Failure
When it fires	Task ran but exceeded the sla timedelta from DAG start	Task threw an uncaught exception
Callback function	sla_miss_callback on the DAG	on_failure_callback on the task or default_args
Task state after event	Can still be running or succeed afterward	Task state becomes FAILED; all downstream become SKIPPED
Business impact	Data will be late — SLA breached even if pipeline completes	Data may be missing or wrong — pipeline stopped
Recovery	Monitor and alert; investigate if repeated	Fix root cause; clear-and-rerun from failed task

IMPORTANT

Always configure BOTH `sla_miss_callback` on the DAG AND `on_failure_callback` on tasks. They are independent. A slow task

CHAPTER 4

SLA and Incident Management

Priority levels · RCA · Communication · Monitoring

Priority Levels

Priority	Definition	Response	Resolution	Examples
P1 Critical	Complete outage — dashboards broken, pipeline down	15 min	2 hours	Snowflake unreachable; all dashboards show no data; ETL not run in 6 hours
P2 High	SLA breach; major data quality; key pipeline very late	30 min	4 hours	Pipeline 3 hours late; wrong revenue in dashboard; dbt failing
P3 Medium	Degraded performance; minor data quality; non-critical delay	4 hours	24 hours	Single analyst query slow; one report has old data
P4 Low	Enhancement; minor issue; documentation; optimization	24 hours	5 days	Feature request; query optimization; new monitoring alert

Escalation Decision Tree

- Escalate to L3 or Engineering when:
 - Root cause requires a code change in the pipeline — not a configuration fix
 - Snowflake account-level issue — requires Snowflake Support engagement
 - More than 50% of SLA resolution window has elapsed with no progress
 - Data corruption or data loss is suspected
 - Security breach or unauthorized access has been detected
- Handle at L2 when:
 - Access or permission issue — grants, roles, masking policies
 - Configuration issue — warehouse size, AUTO_SUSPEND, task schedule
 - Pipeline recovery — REFRESH pipe, clear-and-rerun task, resume stream
 - Known error with an existing runbook fix

RCA Structure — Eight Required Sections

Section	Content Required
1. Incident Summary	Three sentences: what happened, when it started, total business impact
2. Timeline	Chronological list with timestamps: detected, acknowledged, diagnosed, fixed, resolved
3. Root Cause	The specific technical reason — be precise, not generic
4. Contributing Factors	What conditions made this incident possible or worse
5. Impact	Number of users affected, data staleness duration, SLA breach details
6. Resolution	Exactly what was done step by step to fix the issue
7. Action Items	Each with: owner name, due date, and which recurrence it prevents

8. Lessons Learned	Process or monitoring change that comes from this incident
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Stakeholder Communication Templates

TEMPLATE 1 — Initial Acknowledgement

Subject: [INVESTIGATING] Data Pipeline Delay

Hi [stakeholder],

We are aware that [describe issue in plain English — e.g. morning sales dashboard is showing yesterday's data].
Our team is actively investigating.

TEMPLATE 2 — Resolution Notice

Subject: [RESOLVED] Data Pipeline Delay — All Systems Normal

Hi [stakeholder],

The data pipeline issue has been fully resolved as of [time].

What happened: [1-sentence plain English explanation]

CHAPTER 5

Behavioral Questions

STAR framework · Top 10 questions · Escalation

The STAR Framework for L2

Every behavioral answer should follow STAR. In L2 support, interviewers weight the Action section most heavily — they want to know exactly what YOU did, step by step, not what 'we' did or what 'typically' happens.

Component	Weight	What to Cover
Situation	15%	What was broken, who was impacted, what was the severity
Task	15%	What your specific role was — L2, on-call, incident commander
Action	50%	Exact tools used, exact commands run, exact communication sent every 30 min
Result	20%	Resolution time, MTTR vs SLA, preventive action, RCA delivered

Top 10 Behavioral Questions with Answer Guides

Q1. Tell me about a time you handled a P1 incident end-to-end.

HARD

Answer:

Mention: severity was P1 because dashboards were showing no data (total business impact). Walk through the exact diagnosis sequence: first checked SYSTEM_PIPE_STATUS, then Query Profile, then ACCOUNT_USAGE.

Show stakeholder communication: acknowledged in under 15 minutes, sent updates every 15 minutes, wrote RCA within 24 hours.

Show prevention: added a monitoring alert so future outages are detected faster.

Q2. How do you decide when to escalate versus handle a ticket yourself?

MEDIUM

Answer:

Use a four-question checklist:

1. Is the root cause outside my access or technical expertise?
2. Have I spent more than 50% of the SLA resolution window without progress?
3. Does it require a code change rather than a configuration fix?
4. Is there data corruption or a security risk involved?

If any answer is yes — escalate immediately while continuing to investigate in parallel.

Q3. How would you explain a pipeline failure to a non-technical stakeholder?

MEDIUM

Answer:

Template: 'Our automated data delivery process experienced a technical issue at TIME. This means SPECIFIC DATA is currently HOURS old instead of being updated as expected. No data has been lost — the

system is catching up automatically. Your dashboards will show current data by ETA. We are adding monitoring so we detect this faster next time.'

Never say: 'The Snowpipe SQS notification disconnected from the S3 event integration.' Always translate: 'The connection between our data loader and cloud storage was interrupted.'

Q4. Describe a time you improved a process after a recurring incident.**MEDIUM****Answer:**

Key points: identify the recurring pattern (ticket volume, same root cause), what you proposed (runbook, automated alert, code fix), how you got buy-in from the team and engineering, measurable result (tickets reduced, MTTR improved).

Q5. How do you prioritize when you have 5 open tickets and a P1 comes in?**HARD****Answer:**

P1 takes immediate priority — acknowledge within 15 minutes regardless of current load.

For existing tickets: quickly assess which are closest to SLA breach. Hand off the most critical non-P1 to a colleague if available. Communicate to waiting ticket owners that you are handling a critical incident.

Document P1 actions in real-time so another engineer can take over if needed.

Quick Reference — The L2 Answer Formula**FOR EVERY TECHNICAL SUPPORT QUESTION**

1. **DIAGNOSE** — 'First I would check...' (name specific tool or query)
2. **ROOT CAUSE** — 'The most likely cause is... because...'
3. **FIX** — 'I would resolve this by...' (specific command or step)
4. **COMMUNICATE** — 'I would update the stakeholder saying...' (plain English)
5. **PREVENT** — 'To prevent recurrence, I would add...' (monitoring or runbook)

One-Line Answers — Know These Cold

Topic	The One Thing
Snowflake slow query	Query Profile — check partition scan ratio, spill, result cache bypass
Snowflake access error	4 grants required: DB usage + Schema usage + Table select + WH usage
Snowpipe not loading	SYSTEM dollar PIPE_STATUS first — check executionState and notificationChannelStatus
dbt model failed	Compilation vs run-time. Copy compiled SQL to Snowflake to get the actual error.
dbt test failed	SELECT the failing rows manually. Fix source data or adjust test severity.
Airflow task stuck	Check state: running=slow, queued=pool/concurrency, zombie=worker died
Airflow SLA miss	SLA miss is NOT failure. Task ran slowly. Check upstream slowness, warehouse size, pool.
P1 incident	Acknowledge in 15 min. Update every 15-30 min. Escalate at 50% of SLA window.
When to escalate	Code change required / 50% SLA elapsed / data corruption / security breach
Stakeholder update	What happened (plain English) + current impact (specific) + ETA (specific time)